SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. CALM: (A) disable (B) disapprove (C) dishearten (D) disoblige (E) disturb

2. AMATEUR: (A) spectator (B) professional (C) genius (D) outsider (E) entrepreneur

3. SPIRITUAL: (A) effervescent (B) significant (C) adaptable (D) felicitous (E) worldly

4. EXERTION: (A) innocence (B) inactivity (C) indecisiveness (D) incompetence (E) indifference

5. KNOTTY: (A) youthful (B) newfangled (C) moist (D) easy to solve (E) known to be true

6. LUNGE: (A) abort (B) intercept (C) resist (D) revert (E) recoil

7. NULLIFY: (A) use wisely (B) take seriously (C) perform easily (D) make effective (E) cause annoyance

8. CHAMPION: (A) recline (B) struggle (C) regulate (D) persecute (E) accompany

9. CAPRICIOUS: (A) tardy (B) vindictive (C) predictable (D) alluring (E) erroneous

10. LITHE: (A) restrained (B) inflexible (C) distraught (D) complacent (E) imprecise

11. ELUCIDATE: (A) disconnect (B) avoid (C) confuse (D) appear (E) fasten

12. PERIPHERAL: (A) essential (B) fatuous (C) discursive (D) sectarian (E) persuasive

13. EXEMPLARY: (A) vulnerable (B) uncertain (C) foreign (D) contemptible (E) heedless

14. DISAPPROBATION: (A) coercion (B) sanction (C) negligence (D) exemption (E) restriction

15. CALLOW: (A) barren (B) brave (C) mature (D) strict (E) generous

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. The instructor added the restriction that all projects had to be ----: no student could research an area that had been investigated previously by anyone else.

(A) acceptable (B) useful (C) extensive (D) authoritative (E) original

17. Just as congestion plagues every important highway, so it ---- the streets of every city.

(A) delimits (B) delays (C) clogs (D) obviates (E) destroys

18. Some microbiologists believe that the attempt to kill the microbes that live on or in our bodies is a mistake and that the use of antiseptics cannot always be ----.

(A) justified (B) spurious (C) rejected (D) estimated (E) anticipated

19. Although we know that our adversaries' pace is ---- compared to our own, we would be foolish to be ----.

(A) languid. .vigilant (B) torpid. .complacent (C) rapid. .callous (D) accelerated. .prudent (E) dilatory. .pessimistic

20. The ---- theory of the nature of art is vulnerable to the ---- that much art makes no attempt to create likenesses of objects.

(A) abstract. .rebuttal
(B) aesthetic. .criticism
(C) expression. .challenge
(D) modern. .hypothesis
(E) imitation. .objection

GO ON TO THE NEXT PAGE

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Princeton, N.J. 08541

1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

African art is one of the fountainheads of modernist style in contemporary art; this happy accident has saved what otherwise might have been a lost cultural heritage for Afro-Americans. Lack of continuity with traditional African culture and its arts had left premodern Afro-American artists with no advantage, other than sentimental, in either its use or understanding. Their modern counterparts, however, cannot escape the African influence that has since become an integral part of modern art idioms. The tradition of African art, if it is properly understood and assimilated, should and can have even greater influence on the work of black artists today.

In the 1920's, when many young black artists first became aware of this heritage, a sudden and hectic interest flared up. The artistic results, however, suggest that these artists had a relatively superficial understanding; African art could yield little through direct imitation. Having already exerted its primary influence on modern art, to bear fresh fruit, it must be restudied and approached not so much through the mannerisms of its style as through its basic principles. The latter still have rich inspiration to offer to any artist, but especially to artists who may sense a close spiritual kinship with African art. Sound analysis of African art, then, should teach lessons of vigorous simplicity and vitality. On the cultural side, these lessons are those of art for use, of art with a sound and vital rooting in its own cultural soil, and of art not for restricted audiences but for wide popular appreciation. On the technical side, there is the even more important lesson of originality itself, and of the subordination of technique to complete harmony with the artistic idea. For black artists, particularly, there should come from the study of their own ancient and distinctive art tradition a sense of artistic independence and freedom from timid cultural indebtedness.

African art belongs in two native categories–ritualistic and craft art, both more basic than our idea of "fine art." Most African masks and statuettes serve a ceremonial function, and their primary significance is symbolic. They serve to evoke a mood, to enhance the dignity and authority of priest, witch doctor, or chieftain, to symbolize the ancestors or lineage and supernatural attributes of the gods. As talismans, they often serve additionally to control the supernatural spirits by sympathetic magic. Yet closely allied with all this is an appreciation of their artistic beauty and technical skill.

21. The passage primarily concerns the
    (A) symbolic importance of ritualistic African art
    (B) significance of African art for Afro-American artists today
    (C) techniques employed in the creation of original African art
    (D) way African art was used in Africa
    (E) differences between African art and the work of black artists in America

22. According to the passage, one of the cultural lessons taught by African art is that art should
    (A) combine many divergent points of view
    (B) reflect the personality of its creator
    (C) attempt to reform social injustice
    (D) appeal to great numbers of people
    (E) intrigue its audience with its complexity

23. The author's attitude toward African art is primarily one of
    (A) puzzlement
    (B) skepticism
    (C) nostalgia
    (D) exasperation
    (E) admiration

24. Which of the following titles best summarizes the content of the passage?
    (A) How African Art Developed Its Unique Style
    (B) What African Art Can Teach
    (C) Techniques Used by Black Artists
    (D) The Future of Art in Africa
    (E) The Universal Appeal of Art

25. According to the information in the passage, an artist inspired by the basic principles of African art would most likely have produced which of the following?
    (A) A detailed copy of an ancient statuette
    (B) An original carving of a dancer in a technique that fully complements the image
    (C) An abstract sculpture of a warrior comprehensible only to a few
    (D) A complex panorama representing the history of blacks in America
    (E) A realistic reproduction of an African mask

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

GO ON TO THE NEXT PAGE

(This passage was written in 1929.)

Suppose that a rod is moving at very high speed. At first it is oriented perpendicular to its line of motion. Then it is turned through a right angle so that it is along the line of motion. The rod contracts. This contraction, known as the FitzGerald contraction, is exceedingly small under ordinary circumstances. The size of the contraction does not depend at all on the material of the rod, but only on its speed.

It may seem surprising that the dimensions of a moving rod can be altered merely by pointing it different ways. But what rod is being considered? If the rod is thought of as continuous substance, extending in space because it is the nature of substance to occupy space, then there seems to be no valid cause for a change of dimensions. But the rod is really a swarm of electrical particles moving about and widely separated from one another. The marvel is that such a swarm should tend to preserve any definite extension. The particles, however, exert electrical forces on one another, and the volume they fill corresponds to a balance between the forces drawing them together and the diverse motions tending to spread them apart. When the rod is set in motion, these electrical forces change.

There is really nothing mysterious about the FitzGerald contraction. It is an entirely natural property of particles that are held in a delicate balance by electromagnetic forces and that occupy space by buffeting away anything that tries to enter. Or you may look at it this way: your expectation that the rod will keep its original length presupposes that it receives no unfair treatment and is not subjected to any new stresses. But a rod in motion is subjected to a new magnetic stress as a necessary consequence of its own electrical constitution; and under this stress the contraction occurs. Perhaps you will think that if the rod were rigid enough it might be able to resist the compressing force. That is not so; the FitzGerald contraction is the same for a rod of steel and for a rod of rubber. The degree of rigidity and the compressing stress are bound up in the electrical constitution of the rod in such a way that if the degree of rigidity is great so also is the stress. It is necessary to rid our minds of the idea that this failure to keep a constant length is an imperfection of the rod. The FitzGerald contraction is not an imperfection but a fixed and characteristic property of matter, like inertia.

26. The major purpose of the author is to

(A) demonstrate a method
(B) compare evidence
(C) raise a question
(D) debate a point
(E) describe a phenomenon

27. The author feels that the occurrence of the FitzGerald contraction in a rod is less surprising than the fact that the

(A) rod is moving at high speed
(B) degree of the contraction is small
(C) rod maintains any definite shape
(D) particles in the rod are held together by electromagnetic forces
(E) length of the rod decreases when the rod is perpendicular to its line of motion

28. According to the passage, the occurrence of the FitzGerald contraction is most directly related to

(A) the stress exerted on the particles in an object
(B) the material from which an object is made
(C) the direction in which an object is moving
(D) the forces that the particles in an object exert on other objects
(E) an imperfection in the structure of an object

29. It can be inferred that the author considers the occurrence of the FitzGerald contraction to be

(A) unexplainable in terms of present scientific knowledge
(B) unexpected in view of common-sense notions about physical objects
(C) irrelevant to the consideration of other problems in physics
(D) undetectable by any method now known
(E) inconsistent with other physical properties of matter

30. When the author refers to the idea that a solid rod is "continuous substance" (lines 13-14), he implies that this idea is which of the following?

I. A common conception of the nature of solid matter
II. A concept that is not particularly useful for explaining the FitzGerald contraction
III. An accurate description of some kinds of matter

(A) I only (B) III only (C) I and II only
(D) II and III only (E) I, II, and III

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. In order to make the best use of available human resources, we must first ---- and then ---- human talents.

(A) educate. .equalize (B) decompose. .rebuild
(C) revitalize. .discern (D) discover. .develop
(E) produce. .accrue

32. Eastlake is a colorful and ---- commentator, equally at home in high art, folk rock, theology, and economics.

(A) obtuse (B) austere (C) insipid
(D) versatile (E) provincial

33. Ms. Wilton urged patience and ---- in dealing with the protesters rather than the unyielding attitude the administration had adopted.

(A) obstinacy (B) desperation (C) arrogance
(D) compromise (E) retaliation

34. His subjection bred a longing for self-direction, all the stronger for his underlying sense of the ---- of ever achieving it.

(A) irrevocability (B) impossibility
(C) inevitability (D) credibility
(E) predictability

35. Unfortunately, certain aspects of democratic government sometimes put pressure on politicians to take the easy way out, allowing ---- to crowd out ----.

(A) exigencies. .necessities
(B) immediacies. .ultimates
(C) responsibilities. .privileges
(D) principles. .practicalities
(E) issues. .problems

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. HANGAR : AIRPLANE :: (A) stable : horse
(B) canal : ship (C) lobby : administrator
(D) junkyard : automobile (E) bed : river

37. VANE : WIND DIRECTION ::
(A) thermometer : mercury
(B) speedometer : pedal
(C) hourglass : sand
(D) barometer : heat
(E) sundial : time

38. ANONYMOUS : NAME ::
(A) colorful : hue (B) enormous : size
(C) shapeless : form (D) brief : significance
(E) precise : measurement

39. SYMPATHY : EMOTION ::
(A) sorrow : happiness
(B) friendship : relationship
(C) pride : punishment
(D) criticism : guilt
(E) harshness : helpfulness

40. HABIT : CUSTOMARY ::
(A) response : coordinated
(B) impulse : corrective
(C) reflex : involuntary
(D) concern : commendatory
(E) performance : descriptive

41. THREAT : HOSTILITY ::
(A) plea : clemency
(B) promise : benevolence
(C) lampoon : praise
(D) capitulation : malice
(E) compliment : admiration

42. MAGNET : IRON :: (A) tank : fluid
(B) hook : net (C) sunlight : plant
(D) spray : tree (E) flame : bird

43. ARDENT : INTERESTED ::
(A) depressed : cheerful (B) bored : curious
(C) shy : listless (D) incensed : annoyed
(E) wise : detached

44. VOGUE : ACCEPTED :: (A) ferment : coalesced
(B) transition : unchanged (C) jeopardy : voided
(D) abundance : desired (E) disrepute : shunned

45. HYPERBOLE : LANGUAGE ::
(A) prodigality : spending
(B) whimsicality : poetry
(C) idiom : slang
(D) repetition : behavior
(E) ambition : working

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

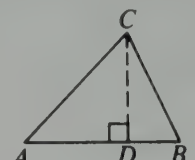
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

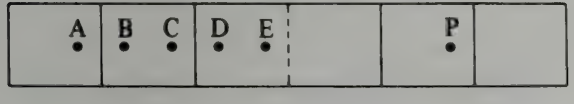
(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $22 \times 3 \times Q = 6$, then $Q =$

(A) $\frac{1}{11}$ (B) $\frac{1}{10}$ (C) 10 (D) 11 (E) 20



2. All the boxes in the strip above are of equal size. When the strip is folded together along the dotted line, point P is most likely to coincide with point

(A) A (B) B (C) C (D) D (E) E

---

3. How many tenths of a mile will a car travel on a 100-mile trip?

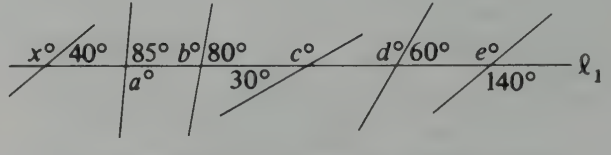
(A) 1,000

(B) 100

(C) 10

(D) 1

(E) $\frac{1}{10}$



4. In the figure above, six segments intersect line ℓ_1. Which of the degree measures, a, b, c, d, or e, is equal to x?

(A) a (B) b (C) c (D) d (E) e

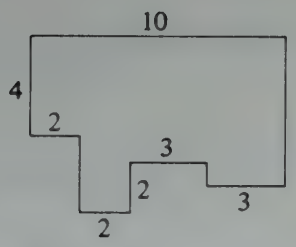
---

5. A grocer has 100 apples, 100 oranges, and 100 pears. If he packs 1 apple, 2 oranges, and 1 pear in a bag, then the maximum number of bags he can fill in this manner is

(A) 20
(B) 25
(C) 50
(D) 75
(E) 100

GO ON TO THE NEXT PAGE

2



Note: Figure not drawn to scale.

6. What is the perimeter of the figure above?

(A) 15 (B) 20 (C) 26 (D) 32
(E) It cannot be determined from the information given.

---

7. $(45)^2 + 2(45)(55) + (55)^2 =$

(A) 5,050
(B) 9,100
(C) 9,900
(D) 10,000
(E) 14,950

8. Add $8x$ to $2x$ and then subtract 5 from the sum. If x is a positive integer, the result must be an integer multiple of

(A) 2
(B) 5
(C) 8
(D) 10
(E) 15

9. A line segment containing the points (0, 0) and (12, 8) will also contain the point

(A) (2, 3)
(B) (2, 4)
(C) (3, 2)
(D) (3, 4)
(E) (4, 2)

10. If $x < 2$, which of the following is NOT true?

(A) $2 + x < 2 + 2$

(B) $x - 2 < 2 - 2$

(C) $x(2) < 2(2)$

(D) $2 - x < 2 - 2$

(E) $\frac{x}{2} < \frac{2}{2}$

11. If $2\frac{3}{8} = 1 + \frac{x}{24}$, then $x =$

(A) 9
(B) 27
(C) 30
(D) 33
(E) 57

$$\begin{array}{r} 2\,3\,5 \\ \times\ 4\,\triangle\,7 \\ \hline 1\,6\,4\,5 \\ 1\,\square\,1\,0 \\ 9\,4\,0 \\ \hline 1\,0\,\otimes\,7\,4\,5 \end{array}$$

12. In the multiplication problem above, if $\triangle$, $\square$, and $\otimes$ represent digits, what digit does $\square$ represent?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

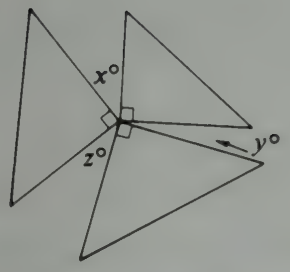


13. Each jar above contains 6 marbles. What is the LEAST number of marbles that must be transferred to make the ratio

marbles in X : marbles in Y : marbles in Z = 3 : 2 : 1 ?

(A) 6 (B) 5 (C) 4 (D) 3 (E) 2

GO ON TO THE NEXT PAGE



14. In the figure above, three right triangles have a common vertex. $x + y + z =$

(A) 90 (B) 120 (C) 135
(D) 180 (E) 270

---

15. If $x \neq 0$, which value(s) of p satisfy the equation $px = p^3x$?

(A) 0 only (B) 1 only (C) 1 or −1 only
(D) 0 or 1 only (E) 0, 1, or −1

16. Sue ate $\frac{1}{3}$ of a sandwich at noon and then $\frac{1}{2}$ of the remainder at supper. What part of the sandwich remained uneaten?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$

17. If the diameter of a bicycle wheel is 0.5 meter, how many meters has the center of the wheel traveled when the wheel has made 3 complete revolutions along a straight road?

(A) $\frac{3\pi}{2}$
(B) 3π
(C) 12π
(D) $\frac{25\pi}{2}$
(E) 25π

18. If S is the sum of 7, 5, 2, 4, and x, what must be the value of x in order for x to equal $\frac{1}{4}S$?

(A) 4
(B) 4.5
(C) 6
(D) 18
(E) 24

	Carson	Greco	Polo	Rand
Carson	0	5	5	1
Greco	5	0	10	4
Polo	5	10	0	6
Rand	1	4	6	0

19. The chart above shows distances in kilometers between four towns that are located along a straight road. Which of the following could be a correct order relationship for these towns along the road?

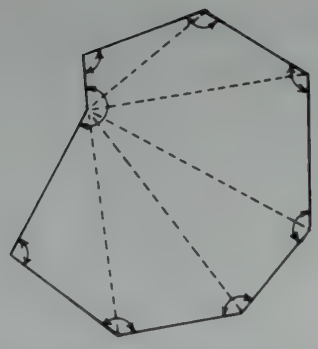
(A) Carson, Polo, Rand, Greco
(B) Carson, Rand, Greco, Polo
(C) Greco, Carson, Rand, Polo
(D) Polo, Carson, Rand, Greco
(E) Rand, Carson, Polo, Greco

---

20. If $3 < x < 7$ and $4 < y < 7$, which of the following best describes $x - y$?

(A) $-4 < x - y < 3$
(B) $0 < x - y < 4$
(C) $3 < x - y < 4$
(D) $3 < x - y < 7$
(E) $4 < x - y < 7$

GO ON TO THE NEXT PAGE



21. In the figure above, what is the average (arithmetic mean) degree measure of the 8 marked angles?

(A) 180 (B) 135 (C) 120 (D) 90
(E) It cannot be determined from the information given.

---

22. For which of the following pairs of numbers is the square of one of the numbers the reciprocal of the other number?

I. 0.25, 2
II. 1, 1
III. 0.5, 4

(A) I only (B) II only (C) III only
(D) I and II only (E) I, II, and III

23. 25 percent of 300 is equal to 7.5 percent of

(A) 10
(B) 75
(C) 90
(D) 100
(E) 1,000

24. What is the volume of a cube with surface area $54x^2$?

(A) $9x^2$ (B) $27x^3$ (C) $81x^2$
(D) $81x^3$ (E) $729x^3$

25. A woman drove to work at an average speed of 40 miles per hour and returned along the same route at 30 miles per hour. If her total traveling time was 1 hour, what was the total number of miles in the round trip?

(A) 30

(B) $30\frac{1}{7}$

(C) $34\frac{2}{7}$

(D) 35

(E) 40

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. HARMONIOUS: (A) appreciative (B) discordant (C) fastidious (D) elementary (E) unworthy

2. PROLONG:
   (A) remain intact
   (B) assume responsibility
   (C) decrease in duration
   (D) fluctuate in value
   (E) reduce in volume

3. SUPERB: (A) wild (B) insecure (C) impractical (D) extremely scarce (E) grossly inferior

4. CURB: (A) demean (B) repudiate (C) improve (D) urge forward (E) deny information

5. DISENTANGLE: (A) prepare (B) lengthen (C) return (D) reprove (E) enmesh

6. RESERVE: (A) compassion (B) irascibility (C) incoherence (D) lack of restraint (E) lack of strength

7. CONFLUENCE: (A) disparagement (B) disturbance (C) diffidence (D) divination (E) divergence

8. INNOCUOUS: (A) durable (B) identical (C) swift (D) dangerous (E) changeable

9. RIBALD: (A) well-liked (B) high-minded (C) tightfisted (D) farsighted (E) aboveboard

10. UPSHOT: (A) initial step (B) quick descent (C) severe punishment (D) complete silence (E) total destruction

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Alice was annoyed that, although Edgar accepted the ---- of her argument, he would not ---- that her conclusion was correct.
   (A) logic. .concede (B) absurdity. .require (C) sequence. .predict (D) existence. .preclude (E) feasibility. .dispute

12. Although the designers have not yet been able to solve the emission problems in the new automobile, their lack of ---- so far cannot be interpreted as final ----.
   (A) continuity. .suspension
   (B) motive. .rejection
   (C) success. .failure
   (D) alternatives. .bafflement
   (E) concern. .defeat

13. A truly ---- historian of science, Meyer neither ---- the abilities of the scientists she presents nor condescends to them.
   (A) unbiased. .scrutinizes
   (B) objective. .inflates
   (C) impressionable. .patronizes
   (D) reverent. .admires
   (E) analytic. .evaluates

GO ON TO THE NEXT PAGE

3

14. By studying ----, psychologists have learned to understand the workings of the healthy mind; similarly, it is in the most ---- instances that the mechanism of many normal processes may be revealed most clearly.

(A) madness. .bizarre
(B) statistics. .unique
(C) genius. .commonplace
(D) animals. .significant
(E) illusion. .real

15. The more knowledge, the fewer ----; wisdom has always been ----.

(A) unknowns. .questionable
(B) certitudes. .elusive
(C) lies. .controversial
(D) benefits. .attainable
(E) aberrations. .subjective

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. STUDIO : ARTIST ::
(A) play : dramatist
(B) cathedral : architect
(C) blackboard : professor
(D) laboratory : chemist
(E) quarry : sculptor

17. RIPPLE : TIDAL WAVE :: (A) breeze : hurricane
(B) blizzard : avalanche (C) valley : earthquake
(D) puddle : downpour (E) rock : waterfall

18. EDUCATION : IGNORANCE ::
(A) book : knowledge
(B) door : entrance
(C) reason : philosophy
(D) medicine : disease
(E) work : recreation

19. BOW : VIOLIN :: (A) music : piano
(B) brass : trumpet (C) note : flute
(D) string : guitar (E) drumstick : drum

20. DUMBFOUND : ASTONISHMENT ::
(A) exasperate : frustration
(B) domineer : rebellion
(C) evaluate : incompetence
(D) err : punishment
(E) repulse : admiration

21. ORBIT : PLANET :: (A) course : ship
(B) rotation : axis (C) destination : plane
(D) tail : comet (E) current : river

22. BENEFACTOR : GIFT :: (A) performer : applause
(B) advisor : counsel (C) contestant : regulation
(D) enthusiast : objectivity (E) inquirer : answer

23. RAZE : BUILDING :: (A) launch : boat
(B) fell : tree (C) adorn : clothing
(D) invade : territory (E) cultivate : garden

24. NERVOUS : EQUANIMITY ::
(A) miserly : greediness
(B) realistic : foresight
(C) pious : grace
(D) gullible : discernment
(E) conscientious : sociability

25. MILL : PENNY :: (A) silver : quarter
(B) currency : nickel (C) dime : dollar
(D) check : cash (E) wallet : money

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

The world of the great forests is a veritable paradise for those who will read its mysteries. Enveloped by towering walls of greenery, like a microbe in a pile carpet, I found never-ending comfort in this safe retreat from the glare of the world. Perhaps these unconventional reactions to an environment hated and feared by many were somehow connected with my upbringing as a city-dweller. Life in great cities is not unlike that in the primeval forests. One passes from one burrow to another along canyons that are only a little lighter, and breadth of vision is everywhere excluded by towering battlements palpitating with the lives hidden behind them. Always there is life around one, but it is hardly visible; imagination must work overtime to formulate its progress, lest one pass forever through the dense crowds utterly alone.

26. The author mentions which of the following as experiences common to visitors in the forests and dwellers in the cities?

I. They are constantly surrounded by other living creatures.
II. Obstructions block their view of vast expanses.
III. They live in fear of attack by hidden enemies.

(A) I only (B) II only (C) I and II only
(D) I and III only (E) I, II, and III

27. The author uses the phrase "towering battlements" (line 13) to describe

(A) ranger towers in a forest
(B) wild animals
(C) remote castles
(D) archaeological excavations
(E) skyscrapers



3

We Americans have steeped ourselves so thoroughly in the practical and the empirical that we take to political theory with reluctance. We do not like to reason things out in advance. We would rather meet situations as they arise and dispose of them with a domestic blend of precedent and rule of thumb. But the situations we now judge vital to our national welfare will not "arise." They will have to be induced; and since democracy draws no distinction between ends and means, if the situations are to conform with democratic ends they will have to be induced by democratic means. The whole prospect calls for meticulous attention to first principles every step of the way. We can no longer afford to say one thing and do another, to talk free enterprise and tolerate monopoly, to maintain an antitrust law on our books and provide more ways of evading than of enforcing it. This is worse than inconsistency. It is an admission of failure that is already being used against democracy with telling effect by its critics and enemies. We can no longer indulge in the cynicism that interprets first principles as mere myths. The opening sentences of *Bulfinch's Mythology* show us the end of that trail: "The religions of ancient Greece and Rome are extinct. The so-called divinities of Olympus have not a single worshiper among living men." If democracy is to survive, we must believe in it, not as myth but as reality.

Line
(5)
(10)
(15)
(20)
(25)

28. Which of the following is the best interpretation of "a domestic blend of precedent and rule of thumb" (lines 6-7)?

(A) A pattern of behavior derived from a study of American history
(B) A combination of tradition and rough calculation based on previous experience
(C) A system of principles based on a study of current American political behavior
(D) A judicious mixture of theory and practical know-how
(E) A blend of the opinions of respected Americans both living and dead

29. In reference to the survival of democracy, the passage conveys primarily a sense of

(A) urgency (B) despair (C) hope
(D) indifference (E) irrationality

30. Which of the following best expresses the meaning of the phrase "interprets first principles as mere myths" (lines 23-24)?

(A) Recognizes that the philosophical basis of democracy comes to us from the remote past
(B) Believes that early conceptions of democracy were erroneous
(C) Considers democratic ideals to be unrealistic
(D) Treats democratic beliefs with religious veneration
(E) Treats political activities as secondary and unimportant

31. The quotation from *Bulfinch's Mythology* is introduced primarily to show that

(A) religious belief is essential to national welfare
(B) beliefs that are not expressed through action will die out
(C) the power of religion is declining
(D) myths cannot survive the attacks of cynics
(E) Greece and Rome failed to survive because they subscribed to the wrong religions

GO ON TO THE NEXT PAGE

"The boat sails on Saturday, Katie, and I want you and the boy to come home with me," said Uncle George.

Line

"I don't want Charlie to leave," Kate said.

(5) "He wants to leave–don't you, Charlie? And I want you to come, too, Katie. It's your home, and you've only got one. The trouble with you, Katie, is that when you were a kid they used to make fun of you in Krasbie, and you just started

(10) running, that's all, and you never stopped."

"I want to go home, Mama," Charlie said. "I'm homesick."

"How could you possibly be homesick for America?" Her voice was sharp. "You've never

(15) seen it. This is your home."

"How do you mean?"

"Your home is with your mother," Kate said.

"There's more to it than that, Mama. I feel strange here all the time. Everybody on the street

(20) is speaking a language that isn't mine."

"You've never even tried to learn the language here."

"Even if I had, it wouldn't make any difference. It would still sound strange. I mean it

(25) would still remind me that it wasn't my language. I just don't understand the people, Mama. I like them all right, but I just don't understand them. I never know what they're going to do next. Sometimes you're homesick, too, I know. I can

(30) tell by the way you look."

"Homesickness is nothing," Kate said angrily. "It is absolutely nothing. Fifty per cent of the people in the world are homesick all the time. But I don't suppose you're old enough to understand.

(35) When you're in one place and long to be in another, it isn't as simple as taking a boat. You don't really long for another country. You long for something in yourself that you don't have or haven't been able to find."

(40) "It's crazy, Katie," said Uncle George. "You come home with me and Charlie. You and Charlie can live in the other half of my house, and I'll have a nice American kitchen put in for you."

"We'll all be happier, Mama," Charlie said.

(45) "We'll all be happier if we have a nice clean house and lots of friends and a nice garden and kitchen and stall shower."

Kate said loudly, "No stall shower or anything else will keep me from wanting to see the world

(50) and the different people who live in it." Then she turned to her son and spoke softly. "You are going to miss this country, Charlie."

32. Uncle George believes that Kate's reluctance to go to America is based on her

(A) scorn for modern American conveniences
(B) unpleasant memories of her life in America
(C) fear that her son's education will suffer there
(D) awareness that she has no other relative living in America
(E) belief that she no longer has anything in common with Americans

33. For which of the following reasons does Charlie say he feels strange in the country in which he is living?

I. Everyone speaks a language other than his own.
II. He does not understand the behavior of the people.
III. He misses the friends he had in America.

(A) I only (B) III only (C) I and II only
(D) I and III only (E) I, II, and III

34. It can be inferred that Kate cites the figure of "Fifty per cent of the people in the world" (lines 32-33) in order to

(A) use an established fact to support her position about travel
(B) convince Charlie that homesickness is good for a person
(C) take advantage of the fact that her son has respect for a majority opinion
(D) elicit sympathy and concern for other people from her son and Uncle George
(E) fabricate a defense of her disinclination to return home

35. Which of the following does Kate appear to value LEAST?

(A) Personal dignity
(B) Her present way of life
(C) Her relationship to her son
(D) Material possessions
(E) Learning about new places and people

36. Which of the following definitions of "home" is closest to Kate's view of home as presented in the passage?

(A) Home is where you feel safest and most secure.
(B) Home is the place that you never want to leave.
(C) Home is the place you are always seeking but can only find in your own soul.
(D) Home is not where you live but where everyone understands you.
(E) Home is the place where you were brought up and where your relatives live.

GO ON TO THE NEXT PAGE

3

What we call science today is not the first attempt to introduce order into nature; it is merely the first attempt to introduce a particular kind of order. In medieval times people lived in an orderly universe, a universe in some ways more orderly than our own. But the order they discovered or imported was obtained by employing interpretative concepts quite different from ours. Their chief concept was that of universal purposefulness. People were at the center of a dependent universe; everything that existed and everything that happened did so in order to be useful to them, even if only in order to teach them moral lessons. Nature was part of a larger scheme in which the central fact was humanity and humanity's immortal destiny. The "material universe" was merely a setting within which a moment of this destiny was being worked out; and phenomena were explained in terms of their supposed purposes. The why of phenomena, not the how of phenomena, was the question that interested the medieval mind.

The basic assumption that was needed to support this medieval position was that nature as a whole is rational, that order can be found in it. The general medieval outlook made this assumption reasonable. Since both nature and human beings had the same author and since nature was designed to forward human destiny, it was not unreasonable to suppose that the workings of nature should proceed in a manner intelligible to the human mind. The modern scientific outlook retains this belief in the rationality of nature, but, since the theory of universal purposefulness has been discarded, contemporary scientists can base their assumptions upon nothing but a pure act of faith. Thus the scientific belief in the rationality of nature is seen to be, historically, an inheritance from a system of thought of which the other terms have been discarded.

37. With which of the following statements would the author be most likely to agree?

(A) The universe is not in fact orderly.
(B) Different concepts can be used to explain the same phenomena.
(C) The medieval concept of order has been disproved.
(D) Modern science should reconsider discarded medieval concepts.
(E) The how of phenomena is of no interpretive value.

38. It can be inferred that the phenomenon of birds gathering food for their young would probably have been interpreted in medieval times as

(A) an instructive example of industriousness
(B) an illustration of imitative action in animal life
(C) an example of the universal operation of instincts
(D) a purposeless and therefore uninteresting phenomenon
(E) a purposeless and therefore unintelligible phenomenon

39. According to the passage, modern science has retained from medieval thought the belief that

(A) nature is part of a larger, eternal scheme
(B) the how of phenomena best reveals the rationality of nature
(C) nature operates according to intelligible principles
(D) all human effort should be directed toward the improvement of human welfare
(E) there are many possible ways to explain natural phenomena

40. On the basis of the passage, modern scientific belief would be in agreement with which of the following statements?

I. Universal order has been verified empirically.
II. The concept of order can exist independent of philosophical justification.
III. The object of science is to explain why things happen.

(A) II only (B) III only (C) I and II only
(D) II and III only (E) I, II, and III

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

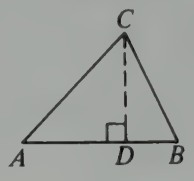
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

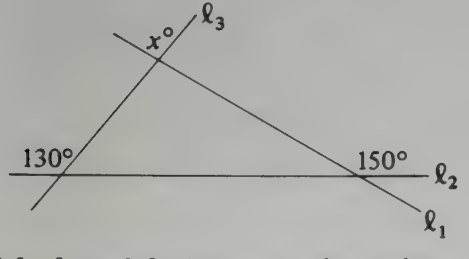
1. $22{,}222 + (5 \times 10^3) =$

(A) 22,722
(B) 25,222
(C) 27,222
(D) 52,222
(E) 72,222

2. A gasoline tank on a certain tractor holds 16 gallons of gasoline. If the tractor requires 7 gallons to plow 3 acres, how many acres can the tractor plow with a tankful of gasoline?

(A) $6\frac{6}{7}$ (B) $7\frac{1}{6}$ (C) $7\frac{1}{3}$

(D) $10\frac{2}{3}$ (E) $37\frac{1}{3}$



3. If ℓ_1, ℓ_2, and ℓ_3 intersect as shown above, then $x =$

(A) 30 (B) 50 (C) 60 (D) 90 (E) 100

4. What is the maximum possible number of digits in the product of two whole numbers each having two digits?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Questions 5-6 refer to the following definition:

Let $D \overset{A}{\underset{C}{\times}} B$ be defined as

$$D \overset{A}{\underset{C}{\times}} B = (A \times C) - (B \times D).$$

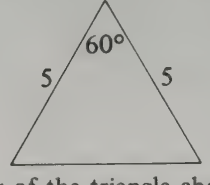
5. If $x \overset{5}{\underset{x}{\times}} 3 = 10$, then $x =$

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

6. $y \overset{x}{\underset{3}{\times}} 2 + 4 \overset{1}{\underset{x}{\times}} y - y \overset{x}{\underset{2}{\times}} 0 =$

(A) $y \overset{x}{\underset{2}{\times}} 1$ (B) $6y \overset{2x}{\underset{1}{\times}} 1$ (C) $4 \overset{6x}{\underset{1}{\times}} 2y$

(D) $5 \overset{2}{\underset{x}{\times}} y$ (E) $3 \overset{3}{\underset{2x}{\times}} 2y$



7. The perimeter of the triangle above is

(A) $5\sqrt{2}$ (B) 10 (C) 12.5
(D) 15 (E) 25

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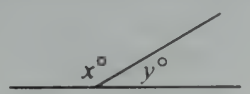
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes:
1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\frac{2}{3} \times N$	$\frac{1}{3} \times 2N$

$$17\overline{)3536}$$

x is the quotient.

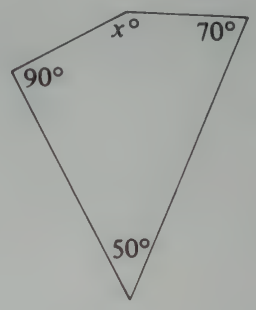
9.	28	x

10.	Average speed, in kilometers per hour, required to travel a distance of 100 kilometers in 2 hours	Average speed, in kilometers per hour, required to travel a distance of 100 kilometers in one-half hour

$n \leqq 12$

11.	$4 + 5 + n$	$4 + 5 + 12$

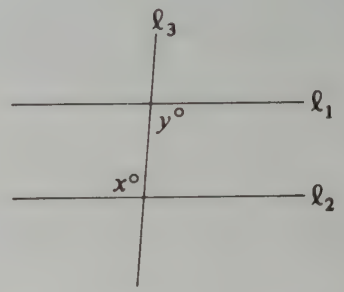
Column A	Column B



12.	x	170°

$x > 0$

13.	$\frac{x^2 + 2x}{x}$	2



Line ℓ_3 intersects parallel lines ℓ_1 and ℓ_2.

14.	$x + y$	$2x$

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

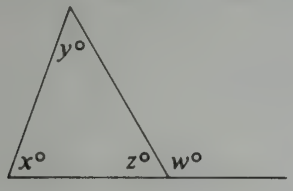
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B



Number Line

15. k | $10 - k$

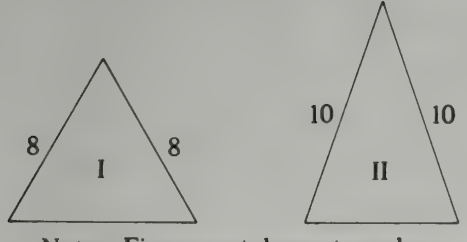


$w = 120$ and $x > y$

16. y | z

$2x < 12 < 3x$ and x is an integer.

17. x | 4



Note: Figures not drawn to scale.

18. Area of triangle I | Area of triangle II

$$\frac{0.8}{0.04} = \frac{0.04}{x}$$

19. x | 0.002

$p > 0 > n$

20. p | $n + p$

	Column A	Column B

21. The number of distinct positive integer divisors of 12 | The number of distinct positive integer divisors of 16

For all real numbers x and y, let $\odot$ be defined as $x \odot y = (x + y)^2$.

22. $p \odot q$ | $p \odot (-q)$

$(x + y)z = 0$
$(x + y) - z = x$

23. y | z

---

24. $\dfrac{\frac{2}{3}}{\frac{3}{2}}$ | 1

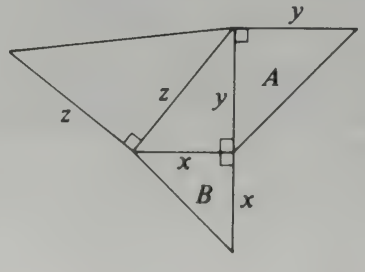
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25. 15% of 2,000 | 2,000% of 15

---

x and y are integers and $x > y > 1$.

26. x^y | y^x



27. Sum of the areas of triangles A and B above | $\dfrac{z^2}{2}$

GO ON TO THE NEXT PAGE

5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. If the average (arithmetic mean) of x and $3x$ is 8, then $x =$

(A) 2
(B) 4
(C) 8
(D) 10
(E) 12

29. If n is an integer greater than 2, which of the following CANNOT be an even integer?

(A) n^2 (B) $n(n-1)$ (C) $n-1$
(D) $3n+1$ (E) $4n+3$



30. In rectangle $ADEH$ above, BG and CF are parallel to AH. Given lengths $AB = 2$, $BC = 1$, $CD = 3$, and the length of DH, not shown, is equal to 10, what is the area of the small rectangle $BCFG$?

(A) 8 (B) 10 (C) 16 (D) 24 (E) 48



31. The LEAST number of trees one would need in order to arrange 4 trees on each line of the plan above is

(A) 5 (B) 10 (C) 14 (D) 15 (E) 20

32. If $\dfrac{1}{1+\frac{1}{x}} = k$, which of the following equals $2k$?

(A) $\dfrac{2}{2+\frac{2}{x}}$ (B) $\dfrac{2}{1+\frac{2}{x}}$ (C) $\dfrac{1}{\frac{1}{2}+\frac{1}{2x}}$

(D) $\dfrac{1}{1+\frac{1}{2x}}$ (E) $\dfrac{1}{2+\frac{1}{2x}}$

33. Given four distinct lines, exactly two of which are parallel, which of the following could be the number of points where at least two of the lines intersect?

I. Three
II. Four
III. Five

(A) I only (B) III only (C) I and II only
(D) I and III only (E) I, II, and III

34. If B is 125 percent of C, then C is what percent of B?

(A) 50%
(B) $66\frac{2}{3}\%$
(C) 75%
(D) 80%
(E) 90%

35. If one side of a ruler is to be marked in $\frac{1}{8}$-inch units and in $\frac{1}{10}$-inch units on the same edge, how many different such marks are needed from the 1-inch mark to the 2-inch mark, including the end points?

(A) 16 (B) 17 (C) 18 (D) 19 (E) 20

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 1Y

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. E	1. B	1. A	1. C
2. B	2. C	2. B	2. A
3. E	3. E	3. A	3. E
4. B	4. D	4. E	4. B
5. D	5. E	5. C	5. D
6. E	6. D	6. E	6. B
7. D	7. E	7. D	7. D
8. D	8. D	8. B	*8. C
9. C	9. B	9. C	*9. B
10. B	10. A	10. D	*10. B
11. C	11. A	11. D	*11. D
12. A	12. C	12. C	*12. B
13. D	13. B	13. D	*13. A
14. B	14. A	14. A	*14. C
15. C	15. B	15. E	*15. B
16. E	16. D	16. C	*16. B
17. C	17. A	17. A	*17. A
18. A	18. D	18. C	*18. D
19. B	19. E	19. D	*19. C
20. E	20. A	20. A	*20. A
21. B	21. A	21. B	*21. A
22. D	22. B	22. E	*22. D
23. E	23. B	23. E	*23. C
24. B	24. D	24. B	*24. B
25. B	25. C	25. C	*25. C
26. E	26. C		*26. D
27. C	27. E		*27. C
28. A	28. B		28. B
29. B	29. A		29. E
30. C	30. C		30. A
31. D	31. B		31. B
32. D	32. B		32. C
33. D	33. C		33. D
34. B	34. E		34. D
35. B	35. D		35. B
36. A	36. C		
37. E	37. B		
38. C	38. A		
39. B	39. C		
40. C	40. A		
41. E			
42. C			
43. D			
44. E			
45. A			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 96. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 96. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 1Y

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 1Y

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	620
84	790		39	460	610
83	780		38	460	600
82	770		37	450	590
81	760		36	440	580
80	760		35	430	570
79	750		34	430	560
78	740		33	420	550
77	740		32	410	550
76	730		31	410	540
75	720		30	400	530
74	710		29	390	520
73	710		28	380	510
72	700		27	380	500
71	690		26	370	490
70	690		25	360	480
69	680		24	360	480
68	670		23	350	470
67	660		22	340	460
66	660		21	330	450
65	650		20	330	440
64	640		19	320	430
63	640		18	310	420
62	630		17	300	410
61	620		16	300	410
60	610	800	15	290	400
59	610	790	14	280	390
58	600	780	13	280	380
57	590	770	12	270	370
56	590	760	11	260	360
55	580	750	10	250	350
54	570	740	9	250	340
53	560	730	8	240	330
52	560	720	7	230	330
51	550	710	6	230	320
50	540	700	5	220	310
49	530	690	4	210	300
48	530	690	3	200	290
47	520	680	2	200	280
46	510	670	1	200	270
45	510	660	0	200	260
44	500	650	– 1	200	260
43	490	640	– 2	200	250
42	480	630	– 3	200	240
41	480	620	– 4	200	230
			– 5	200	220
			– 6	200	210
			– 7 or below	200	200